

Comprehensive Detailed Reflection on Capstone Experience

The most important things I learnt

The Capstone project gave me the opportunity to move beyond theory and really understand how artificial intelligence projects are built in practice. I learnt how important it is to prepare high-quality datasets and how much influence this has on the final system. I created the majority of the training and benchmarking dataset, focusing on pacemaker, ventilator, infusion pump and CPAP devices, and I also prepared an unseen evaluation set so that we could test the model's ability to generalise. I learnt that without careful structuring and consistency, the model could not perform properly, so I designed the dataset in ChatML JSONL format which made it compatible with our fine-tuning pipelines.

I also learnt how to fine-tune large language models efficiently using LoRA in 4-bit mode. This required me to solve tokenizer issues, handle control and pad tokens, and optimise scripts so that they ran in a reproducible way in Google Colab. Through experimentation and step-by-step debugging I managed to reduce evaluation runtime from close to an hour to less than fifteen minutes, which made it possible for the whole team to test and validate results more quickly.

Another important learning was about safety in AI. At first I focused on standard metrics such as precision, recall, F1 and BERTScore, but I realised that these did not capture the real risks of hallucinations in healthcare applications. This led me to design and implement hallucination detection using retrieval-augmented generation with FAISS embeddings and a trust score mechanism. I created a system that classified answers into low, medium and high risk and refined the severity thresholds so that we could reliably measure the factual grounding of outputs. I learnt that in sensitive areas like healthcare, safety and trust are just as important as accuracy.

Finally, I learnt how to evaluate AI systems in a more holistic way. I created an evaluation pipeline that compared pre-fine-tuning, post-fine-tuning and post-fine-tuning with RAG outputs. I collected results in CSVs and prepared comparative plots which showed not only similarity scores but also factual grounding and trust scores. This taught me that a single number cannot explain performance, and that meaningful evaluation requires combining multiple measures and interpreting them carefully.

Did I achieve what I set out to achieve?

When I started, my goal was to fine-tune a model and benchmark it with a dataset. By the end, I had achieved much more. I delivered an end-to-end Medical Device Q&A Guard system that included high-quality datasets, a fine-tuned LLaMA-2 model, an evaluation pipeline with multiple metrics, hallucination detection through trust scores, and two working user interfaces. The Gradio interface allowed us to show responses side by side for pre-fine-tuning, post-fine-tuning and post-fine-tuning with RAG, while the React prototype connected through an API and enabled model switching with trust score overlays. This went far beyond

what I originally expected to complete and showed me that I can manage a full AI development cycle.

The most impressive thing I achieved was

The most impressive part of my work was building the hallucination detection framework and integrating it into the Medical Device Q&A Guard. This was not something I could copy directly from tutorials. I had to research, experiment and combine different techniques to make it work for our domain. By using FAISS embeddings to retrieve relevant context and calculating cosine similarity trust scores, I created a way to measure how reliable an answer was. I then built a classification system that separated responses into low, medium and high risk of hallucination. This achievement stood out because it solved a real challenge in making AI safe for healthcare and it demonstrated innovation and leadership.

Another achievement I am proud of is the user interface development. The Gradio app I built showed results across all three stages of the pipeline and displayed trust scores alongside answers. This made the outputs transparent and easy to understand. I also contributed to the React prototype which was more production-ready and allowed real-time interaction with trust score visualisation. These interfaces made the project accessible to non-technical audiences and showed that I can bridge technical results with practical usability.

The things that helped me most were

One of the things that helped me the most was keeping systematic documentation. After each fine-tuning run or evaluation, I noted down the settings I used, the time taken, and the quality of the outputs. This made it easier to trace issues and reproduce successful runs. I also created CSV logs and progress updates which supported both my own work and team discussions.

Another thing that helped was following a structured debugging process. Instead of getting lost in complex problems, I broke them down into smaller steps, tested changes gradually, and documented what worked. This method helped me solve runtime inefficiencies and tokenisation issues and improved the stability of our fine-tuning pipeline.

I also relied on multiple sources of information such as Hugging Face documentation, GitHub forums, and academic papers. By combining these, I learnt how to apply LoRA in 4-bit precision and how to integrate FAISS with retrieval-augmented generation. This gave me both practical solutions and a deeper understanding of the techniques.

Team collaboration and leadership also played a big role. I mentored a new team member and guided her through dataset preparation and fine-tuning so she could contribute effectively. I coordinated the group submission for Task 6.1P and represented our company Databytes in the trimester induction session with junior students. Presenting our progress during leadership meetings also helped me build confidence and clarity in explaining technical achievements.

I still need to work on the following areas

Although I achieved a lot, I can see areas where I still need to improve. I need to gain more experience with advanced statistical analysis, for example designing experiments with controlled variables, calculating sample sizes, and applying statistical tests when comparing models. This would strengthen the reliability of my evaluations.

I also need to learn more about scalability and deployment in production environments. While I created working demos in Gradio and React, I still need to understand how to manage larger systems that can handle more users, deal with server loads, and integrate monitoring and security.

Another area is domain knowledge. I ensured my system avoided unsafe outputs technically, but collaboration with medical experts would improve the dataset and ensure answers are practical and aligned with healthcare standards.

Finally, I want to deepen my knowledge of model architectures. In this project I successfully applied existing models, but in the future I would like to learn how to customise or design architectures for specialised domains.

This unit will help me in the future

This unit has given me the skills and confidence to work as an AI engineer. By completing the full cycle from dataset to model to evaluation and UI, I now understand how to design and deliver AI systems. The experience with hallucination detection and trust scores is particularly valuable because it prepares me to work in regulated industries like healthcare and finance where reliability and safety are critical.

The leadership experiences I gained will also help me. By mentoring teammates, coordinating submissions, and presenting progress to leadership, I developed skills in communication, teamwork and responsibility. These are essential not just in university projects but in professional environments.

I also learnt how to innovate. Developing the trust score framework showed me how to take an ambiguous challenge and turn it into a practical solution. This mindset will help me in research, development, or even in entrepreneurial projects in the future.

If I did this unit again I would do the following things differently

If I could start again, I would begin with a more detailed risk assessment. Identifying possible runtime bottlenecks, evaluation challenges, and hallucination risks at the start would have saved time later. I would also involve domain experts earlier so that the dataset and evaluation reflected real medical practice more closely.

I would apply systematic hyperparameter optimisation such as grid search or Bayesian optimisation instead of relying mainly on defaults. I would also explore additional medical

datasets to cover more devices and compare against commercial chatbots to give context to our system's performance. These steps would make the project stronger and closer to real-world use.

Other Reflections

Working with limited resources in Colab forced me to be creative. Using LoRA fine-tuning in 4-bit mode turned out to be both efficient and practical. I also realised that AI is a fast-changing field, and keeping up with new techniques like LoRA and RAG requires constant learning.

This project also showed me that AI success depends on more than just code. It required ethical awareness, an understanding of healthcare risks, evaluation methods, and user interface design. Building the system made me more aware of the ethical responsibility AI developers carry, especially in preventing misinformation.

I also learnt the importance of building for others, not just myself. By creating reproducible pipelines, clean datasets, and accessible interfaces, I made sure my work could be used and extended by teammates. The project advanced step by step — from datasets to fine-tuning to evaluation to trust scores and finally to UIs — and this iterative process was key to success.

Closing Reflection

The Capstone experience has transformed the way I see myself as a professional. I started with the aim of fine-tuning a model but ended with a complete Medical Device Q&A Guard system that integrated datasets, fine-tuned pipelines, evaluation frameworks, hallucination detection, and user interfaces. Alongside technical contributions, I took responsibility for mentoring, team coordination, and leadership presentations.

This unit helped me grow as both a technical expert and a leader. I now see myself not just as a programmer, but as an AI practitioner who can design responsible, safe, and practical systems. These are the skills I will carry into my future career in AI engineering and research.